

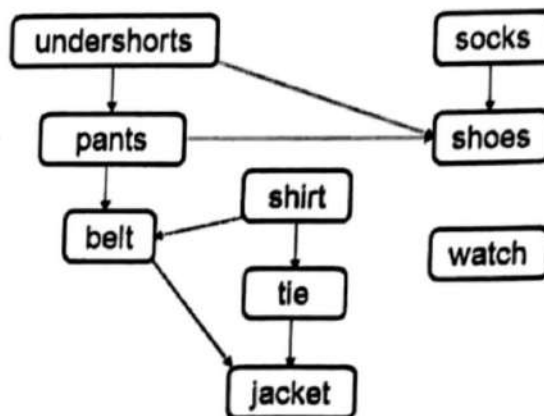
Time: 1 hour 20 minutes

Note: Answer all questions. Figure in the right margin indicate marks.

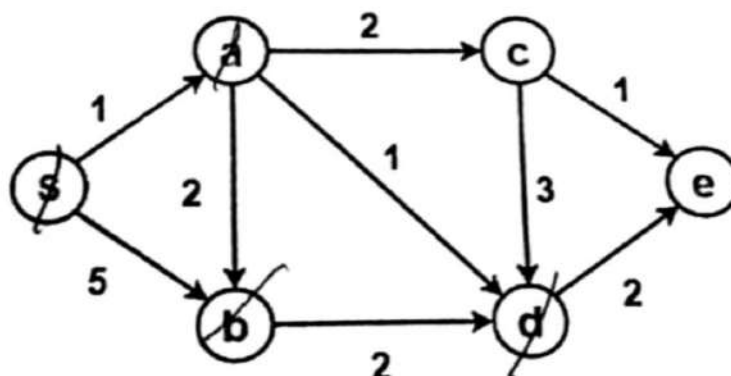
1. Draw a binary search tree using the following numbers. Also, show the linked representation of the BST. 6
 37, 2, 45, 41, 29, 20, 30, 49, 70
2. What is complete binary tree? Find the Huffman tree of the following 7
 information.

Letter	Freq
a	120
d	42
l	42
u	37
c	32
m	24
k	7
z	2

3. What is the minimum spanning tree? Find the discovery and finishing 7
 time of every node in following graph. Also find the forward and cross edge:



4. Explain the condition under which the Bellman-Ford algorithm cannot 10
 determine shortest paths. Find the single source shortest path using
 Dijkstra Algorithm. Assume source is S.

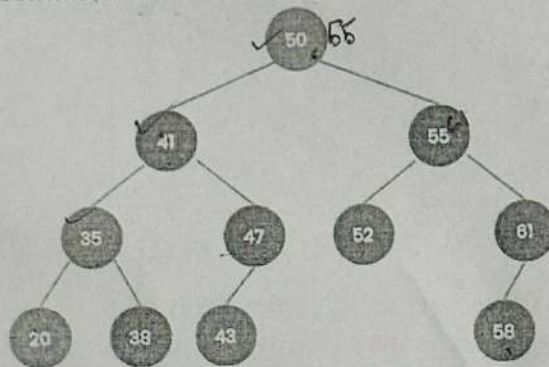


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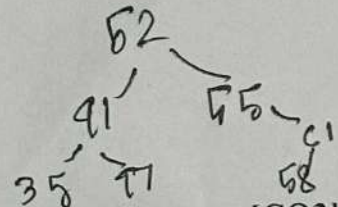
Department of Computer Science and Engineering
 Course Code: CSE 207 Title: Data Structures
 Course Instructor: Manowarul Islam
 Date: 26.05.2025 Spring 2025
 Final Examination Section: 9
 Time: 1 hour 20 minutes Marks: 20

Note: Answer all questions. Figure in the right margin indicate marks.

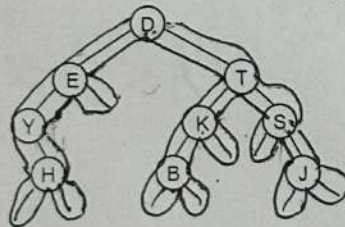
1. State the properties of recursion. Delete 50 from the following tree. Show [CO2]-
 the link representation before and after the delete operation. 6



55 node left right



2. Define complete binary tree. Consider the following tree.
 i. Shows the sequential memory representation of the tree.
 ii. Show the in-order and post-order traversal of the tree.



or

In: H, Y, E, D, B, K, T, S, J,
 Post: H, Y, E, B, K, J, S, T, D

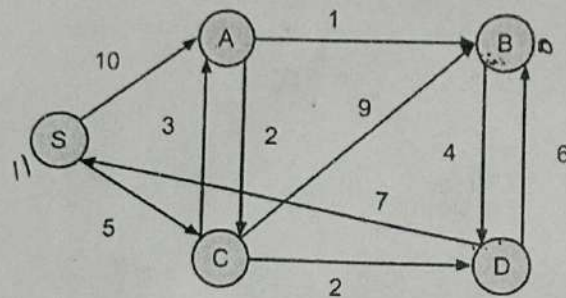
What are the application areas of Huffman coding? Find the code of the following info using Huffman algorithm.

Symbol	Frequency
A	11
B	12
C	13
D	14
E	24
F	26

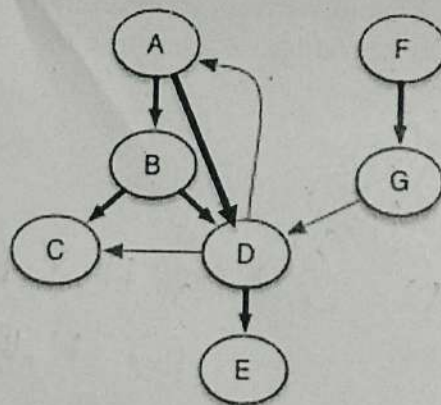
H, Y, E, B, K, J, S, T, D

B	A	B	E	D
D	∞	0	∞	∞
S	∞	∞	∞	∞
E	∞	∞	∞	∞
A	21			

3. What is a negative weight cycle in graph? Find the single source shortest path using Dijkstra Algorithm. Assume source is B. [CO3]-
8



4. Show the d and f values that result from running the Depth-first search (DFS) on the directed graph given below. Also find the forward or cross edge. [CO2]-
7



Define MST? Distinguish between prim's and kruskal's algorithm.

[CO3]-
5

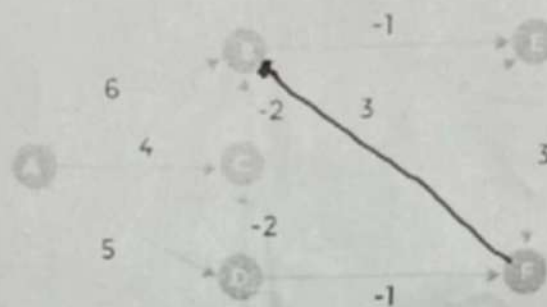
Minimal spanning tree

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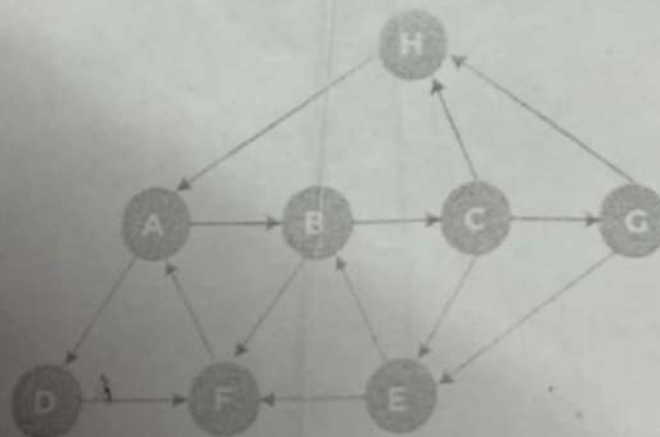
Department of Computer Science and Engineering
 Course Code: CSE 207 Title: Data Structures
 Course Instructor: Manowarul Islam
 Date: 13.09.2023 Summer 2023
 Final Examination Section: 6
 Time: 1 hour 20 minutes Marks: 20

Note: Answer all questions. Figure in the right margin indicate marks.

1. Describe relaxation process with example. Find all pair shortest path of the following graph using bellman-ford algorithm. [CO2]-6.5



2. Differentiate between Prim's and Kruskal's algorithm. [CO2]-2.5
3. State how bellman-ford can identify negative cycle in a graph. [CO3]-3
4. Show the d and f values that result from running the Depth-first search (DFS) on the directed graph given below. Finally find the back edge, cross-edge and forward edge of the graph. Now from your results, state how DFS can detect any cycle in the graph. [CO3]-7.5

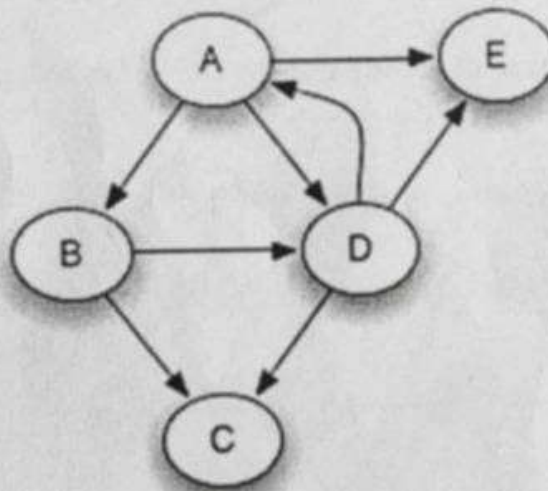


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Department of Computer Science and Engineering
Course Code: CSE 207 Title: Data Structures
Course Instructor: Manowarul Islam
Date: 17.05.2023 Spring 2023
Final Examination Section: 5
Time: 1 hour 20 minutes Marks:20

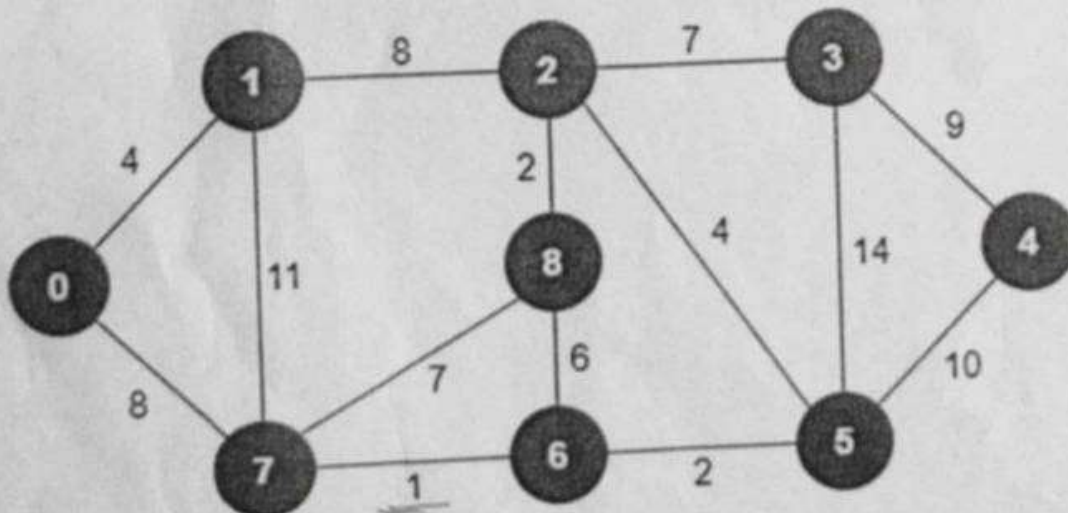
Note: Answer all questions. Figure in the right margin indicate marks.

1. Give an adjacency-list and adjacency-matrix representation of the following graph. Mention some application area of graphs. [CO2]-4

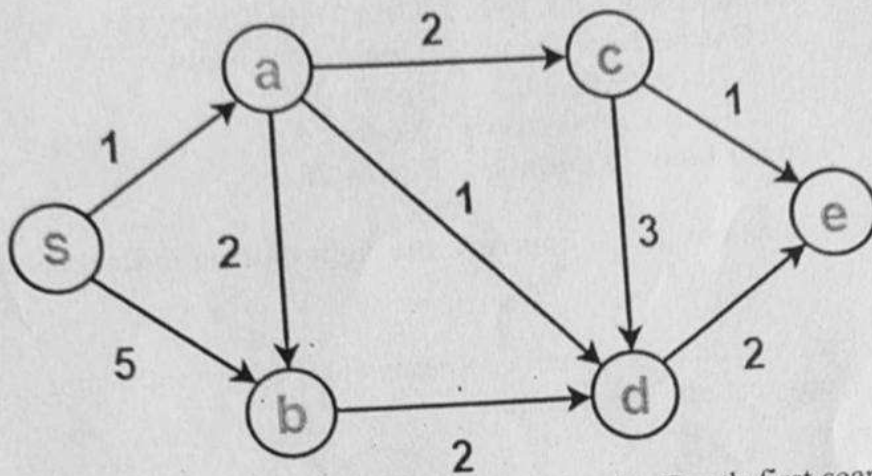


2. How you can detect a cycle in a graph. Explain with example. [CO2]-2.5

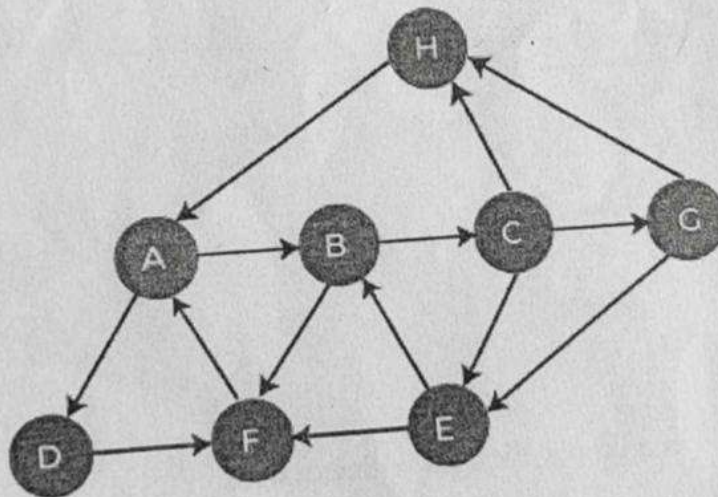
3. Find the MST using prim's algorithm. [CO3]-3.5



4. Find the single source shortest path using Dijkstra Algorithm. Assume source is S. [CO3]-5

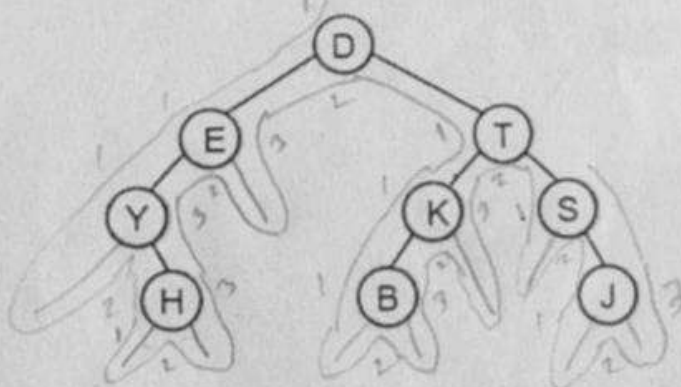


5. Show the d and f values that result from running the Depth-first search (DFS) on the directed graph given below. [CO2]-5



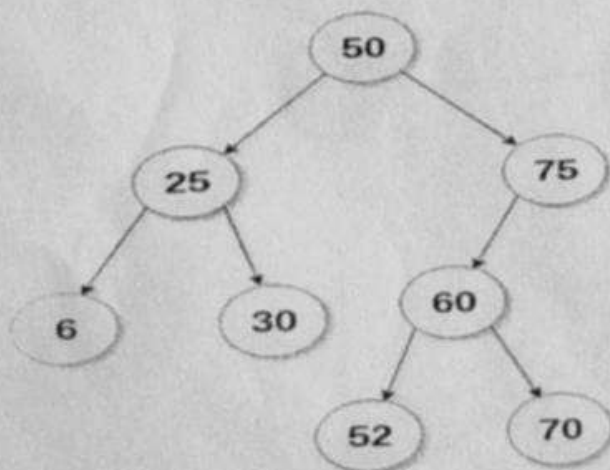
Note: Answer all questions. Figure in the right margin indicate marks.

1. What are the conditions of a well define recursion algorithm. 2
2. State the solution tree for tower of Hanoi problem using N=4. 3
3. What is BST? Draw a binary search tree using the following numbers: 4
 37, 2, 45, 48, 41, 29, 20, 30, 49, 70
4. Consider the following tree. 4
 - i. Shows the sequential memory representation of the tree.
 - ii. Show the in-order and post-order traversal of the tree.



Pre: Y, H, E, D, B, K, T, S, J
 Post: H, Y, E, B, K, J, S, T, D

5. Delete 50 from the following tree. Show the link representation before and after the delete operation. 5



6. Define extended binary tree with example. 2

Complete tree

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Department of Computer Science and Engineering

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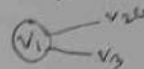
Date: 12.01.2023 Fall 2022

Final Examination Section: 5

Time: 1 hour 20 minutes Marks: 20

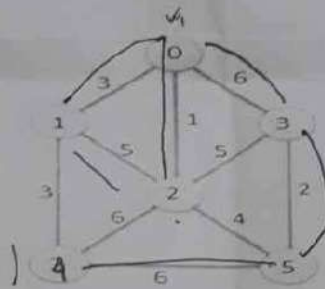
Note: Answer all questions. Figure in the right margin indicate marks.

- ✓ 1. Define connected graph with example. Consider the following graph [CO2]-4
 $G = (V, E): V = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$ and $E = \{v_1v_3, v_1v_5, v_2v_3, v_2v_5, v_3v_5, v_3v_7, v_4v_5, v_4v_7, v_5v_6, v_6v_7\}$ Draw the graph.

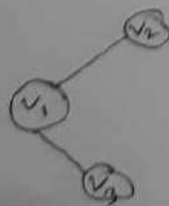
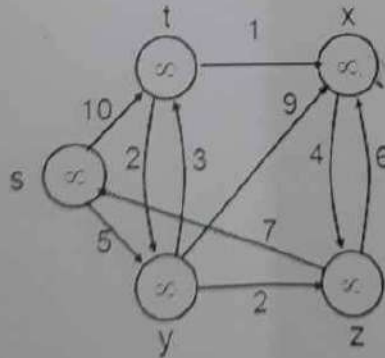


- ✓ 2. State recursive solution for the Tower of Hanoi problem if the number of Discs $N=4$. [CO2]-3

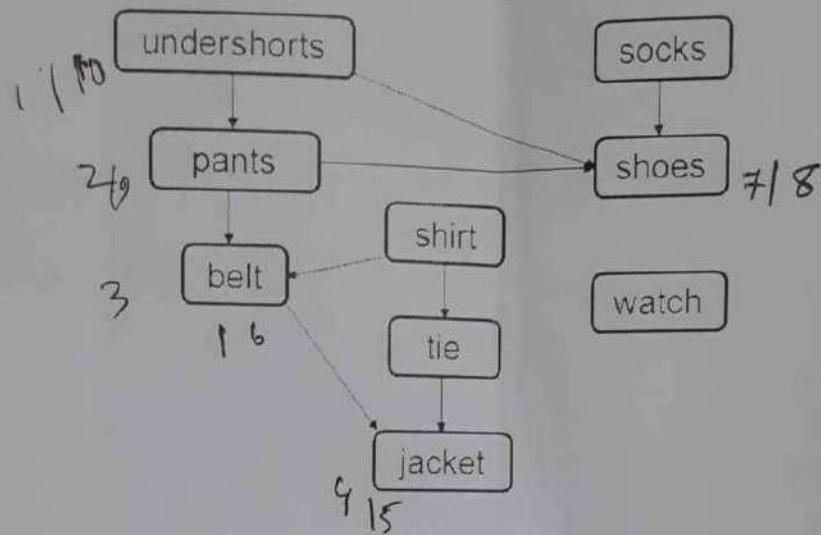
3. Find the MST using prim's algorithm. [CO3]-3



- ✓ 4. Find the single source shortest path using Dijkstra Algorithm. Assume source is S. [CO3]-4



- ✓ 5. Show the d and f values that result from running the Depth-first search (DFS) on the directed graph given below. [C]

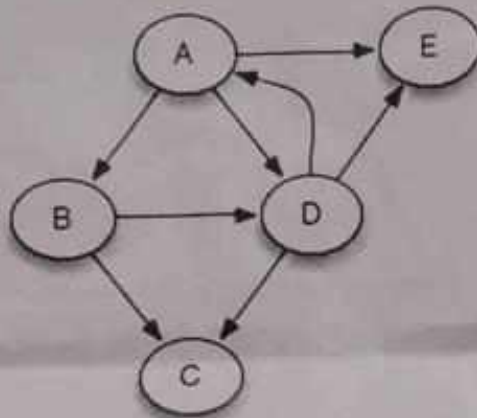


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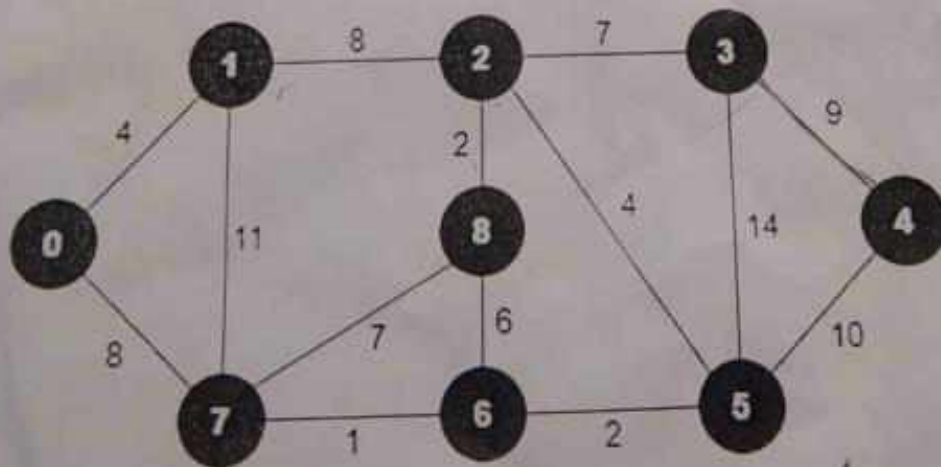
Department of Computer Science and Engineering
Course Code: CSE 207 Title: Data Structures
Course Instructor: Manowarul Islam
Date: 15.09.2022 Summer 2022
Final Examination Section: 5
Time: 1 hour 20 minutes Marks: 20

Note: Answer all questions. Figure in the right margin indicate marks.

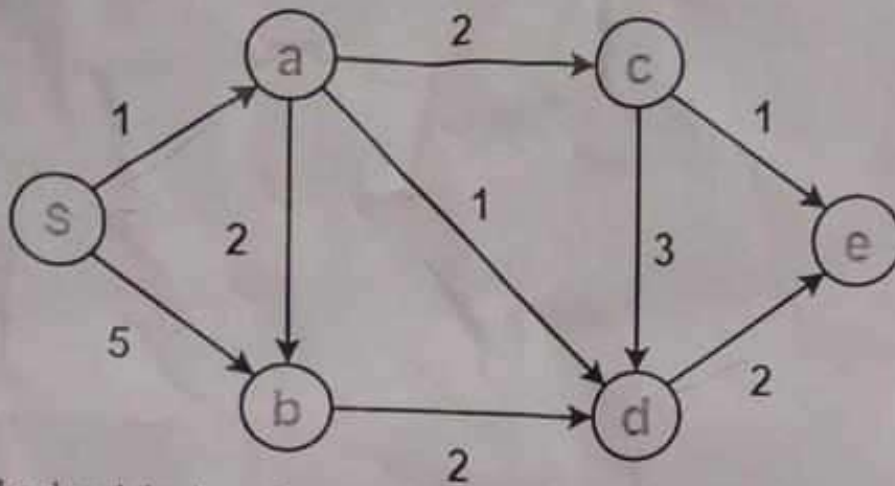
1. Give an adjacency-list and adjacency-matrix representation of the following [CO2]-2.5 graph.



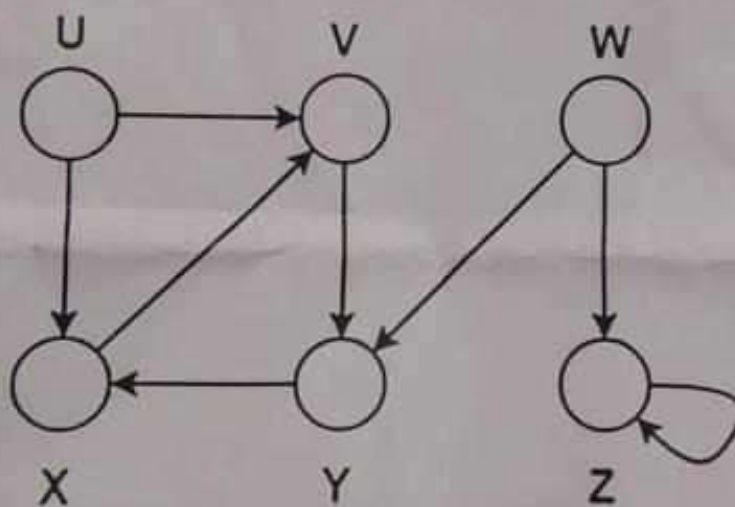
2. What is a Heap? Give example. [CO2]-2.5
3. Find the MST using prim's algorithm. [CO3]-3



4. Find the single source shortest path using Dijkstra Algorithm. Assume source is S. [CO3]-6



5. Show the d and f values that result from running the Depth-first search (DFS) on the directed graph given below (Show each steps clearly.). [CO2]-6

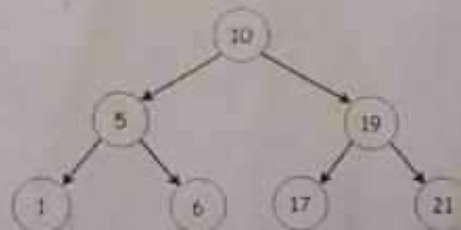


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Department of Computer Science and Engineering
Course Code: CSE 207 Title: Data Structures
Course Instructor: Manowarul Islam
Date: 11.08.2022 Summer 2022
MID-2 Section: 5
Time: 1 hour 20 minutes Marks: 20

Note: Answer all questions. Figure in the right margin indicate marks.

1. Define Extended Binary Tree with example. [CO2]-3
2. What is BST? Draw a binary search tree using the following numbers: [CO2]-4
50, 70, 60, 20, 90, 10, 40, 100
3. Show the linked representation of the following tree: [CO2]-3

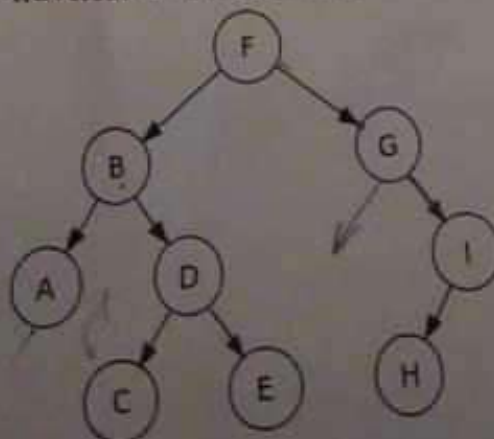


50740

4. How you can delete a node from BST that has one or two children. Explain with example. [CO3]-4
5. Find the Huffman tree of the following information. [CO3]-4

Letter	Freq
e	120
d	42
i	42
u	37
c	32
m	24
b	7
z	2

6. Show the post order traversal of the following tree. [CO3]-2

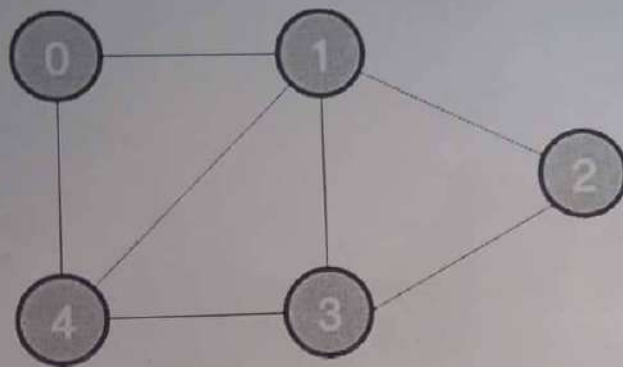


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Department of Computer Science and Engineering
Course Code: CSE 207 Title: Data Structures
Course Instructor: Manowarul Islam
Date: 12.05.2022 Spring 2022
Final Examination Section: 5
Time: 1 hour 20 minutes Marks:20

Note: Answer all questions. Figure in the right margin indicate marks.

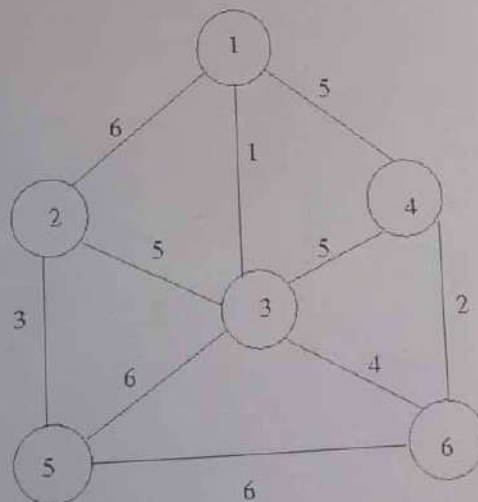
- ✓ 1. Give an adjacency-list and adjacency-matrix representation of the following [CO2]-2.5 graph.



2. What is a Heap? Give example.
3. Find the MST using prim's algorithm.

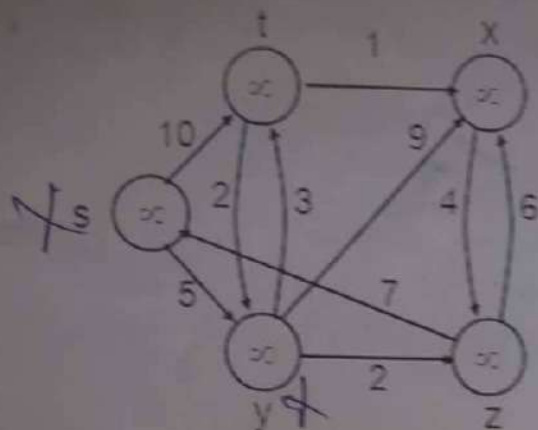
[CO2]-2.5

[CO3]-3

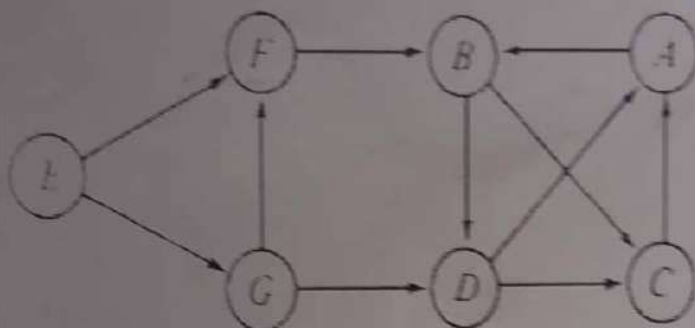


✓

4. Find the single source shortest path using Dijkstra Algorithm. Assume source is S. [CO3]-6

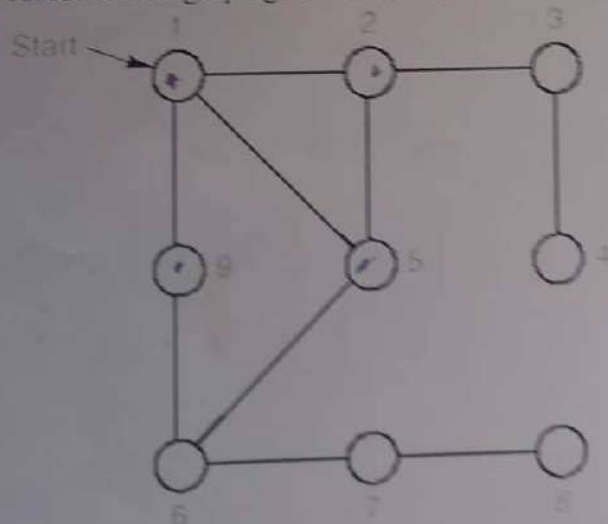


6. Show the d and f values that result from running the Depth-first search (DFS) on the directed graph given below. [CO2]-6



Or

Find the BFS forest for the graph given below: (Start from 1)

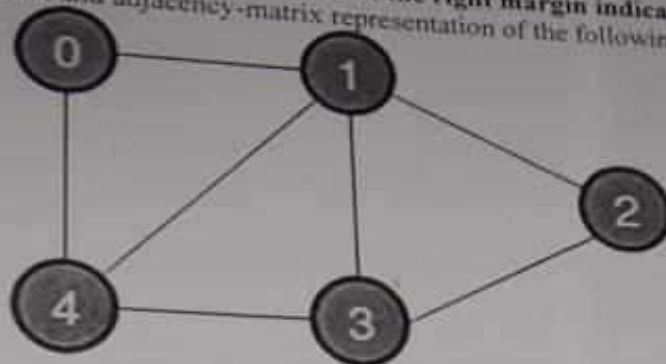


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Department of Computer Science and Engineering
 Course Code: CSE 207 Title: Data Structures
 Course Instructor: Manowarul Islam
 CT 2

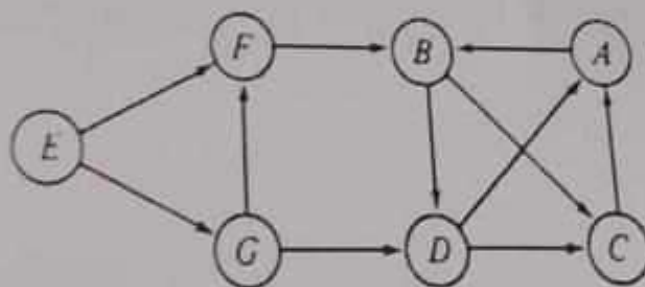
Note: Answer all questions. Figure in the right margin indicate marks.

1. Give an adjacency-list and adjacency-matrix representation of the following graph.



3

2. Show the d and f values that result from running the Depth-first search (DFS) on the directed graph given below.



Or

Find the BFS forest for the graph given below: (Start from 1)

